Multi Channel LoRaWAN Gateway

NLG6X40-IN

Introduction

NLG6X40-IN : High-Performance 8-Channel LoRaWAN Gateway

The **NLG6X40-IN** is a high-performance, industrial-grade gateway designed to anchor large-scale IoT networks. By integrating a robust **8-channel LoRaWAN concentrator** with the proven reliability of the **NXP i.MX6 processor**, it serves as a sophisticated communication hub that maintains secure and seamless data flow, even in the most demanding environments.

Engineered for superior scalability, the NLG6X40-IN ensures consistent performance as your network density increases. It facilitates the reliable collection and transmission of sensor data to the cloud with minimal latency and maximum security. The NLG6X40-IN delivers the stable, long-range connectivity required for modern smart cities and Industrial IoT (IIoT) ecosystems.

Applications

- Smart Utility Metering (Electricity, Gas, Water)
- Industrial Automation & Predictive Maintenance
- Agriculture & Environmental Monitoring
- Smart Street Lighting
- Defence, Military Applications
- Smart Cities

Features

- Long range LoRaWAN wireless data transmission, range up to 15 km (Line of Sight) and over 2 km in dense urban environments
- Support upto 1000 node on single gateway (*based on field scenario this will change)
- Offline data storage of 32 GB
- Wide input supply range (upto 528VAC) with integrated over-voltage protection
- Tamper detection
- External fiber glass antenna with 5 dBi gain
- 10kV surge protection
- Support Gateway with battery backup 4 hrs
- LTE, Ethernet, Wi-Fi (Optional), and GPS (Optional)

Processor	
ARM® Cortex®-A7 Microprocessor IC i.MX6 1 Core, 32-Bit 792MHz 289-MAPBGA	
Operating System	
Yocto Based Linux System	
Storage	
256MB NAND Flash, 512MB DDR3L RAM, 32 GB SD CARD	
LoRaWAN	
Frequency	IN865 (Other Country Option Available. Refer Product Selector List)
Conductive Power	+5 to +27 dBm (Configurable)
Receiver Sensitivity	Down to -137 dBm @SF12, BW 125 kHz
Uplinks channel	8
Antenna	5dBi Fiber Glass Antenna
Cellular	
Cellular bands	LTE FDD – B1/B3/B5#/B7/B8/ B20/B28# GSM - GSM/GPRS/EDGE 900/1800 MHz
SIM card	Nano SIM Tray
Antenna	Inside Enclosure
Wi-Fi (Optional-Refer Product Selector Guide)	
Band	Single Spatial Stream in 2.4 GHz ISM Band
Standards	802.11 b/g/n, 802.3, 802.3u 802.11e-compatible bursting and I standards
Antenna	Inside Enclosure
Ethernet	
1 x IEEE 802.3 10/100M	
USB	
TYPE C for Serial Debug Communication	
Power Supply	
Power supply	85 ~ 528VAC
Power Backup	4 hrs Battery Backup After Power Failure
Power consumption	Average 30W (Approx)
Mechanical Specification	
Size	250 x 210 x 92 mm
Mounting Type	Pole / Wall Mount
Enclosure	Polycarbonate – UV Stabilized, UL94 V0 Rated, IP65 Self Certified, Fire retardant rated case
Environmental Specification	
Operating Temperature	-10°C to 60°C
Humidity	Relative Humidity 15% to 93% Non Condensing
Pre-Compliance Certification	
• ESD (IEC 61000-4-2) • EFT (IEC 61000-4-4) • Fire retardant • Surge (IEC 61000-4-5) • FCC Certified LoRa Concentrator Module	

*LoRa Concentrator Module WPC (In Progress)

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Model Structure

Model Example — **NLG6X40 - IN**

Based SLS Gateway name

NLG6 – SLS Gateway code

X - Feature Selector to Connect Cloud/Server

5 – GSM, Ethernet

7 – GSM, Wi-Fi, and Ethernet

Radio

40 – LoRaWAN

LoRaWAN Region

IN (IN865)

US (US915) US, Canada, Mexico, Bolivia, Peru, Uruguay, Venezuela, Colombia, Ecuador, Panama, Paraguay

EU (EU868) EU, Mauritius, Mozambique, Namibia, Saudi Arabia, South Africa, Turkey, UAE, UK, Finland, Madagascar, Malta, Switzerland, Tanzania

AU (AU915) Australia, Brazil, Argentina, New Zealand, Chile

RU (RU864) Russia

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